6) PO 20%

WAGNER S.C.

$W_n = 0,5 \text{ rad/s}$

$$\xi = \frac{-\ln(0.2)}{(\pi^2 + \ln^2(0.2))^{1/2}} = 0,456$$

$$T_p = \frac{\pi}{\frac{1}{2}\sqrt{1-0,456}} = 7,06s$$

$$T_s = \frac{4}{\xi W_n} = \frac{4}{0,456 \cdot 0,5} = 17,54s$$

$$G(s) = \frac{W_n^2}{s^2 + 2\xi W_s + W_n^2} = \frac{0,25}{s^2 + 0,456s + 0,25}$$

7) $\xi = 0,4$

$W_n = 0,5$

$$G(s) = \frac{W_n^2}{s^2 + 2\xi W_n s + W_n^2} =$$

$$\frac{1/m}{s^2 + \frac{b}{m}s + \frac{K}{m}}$$

$$W_n^2 = \frac{K}{m=1} \rightarrow K = 0,25$$

$$2\xi W_n = \frac{b}{m}$$

$$2 \cdot 0.4 \cdot 0.5 = \frac{b}{1} \qquad b = 0,4$$

$$T_p = \frac{\pi}{0,5(1-0,4^2)^{1/2}} = 7,45s$$

$$T_s = \frac{4}{\xi W_n} = \frac{1}{0,05} = 20s$$

$$PO = 100e^{-\frac{\xi\pi}{\sqrt{1-\xi^2}}} = 100e^{-0,49} = 22,53\%$$

8) $G(s) = \dfrac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2} = \dfrac{2{,}37K}{s^2 + 9.5s + 2.37}$

$PO = 20\% \rightarrow \zeta = \dfrac{-\ln(20\%/100)}{\underbrace{(\pi^2 + \ln^2(20\%/100))^{1/2}}_{2{,}59}} = \underline{3{,}1}$

$t_s = 1s \rightarrow \omega_n = \dfrac{4}{t_s\zeta} = 1{,}54 \text{ rad/s}$

$G_2(s) = \dfrac{as}{s^2 + ds + d} = \dfrac{g}{1 + gh}$

$\text{step}(G_2) \equiv \text{step}(G_c)$ ✓

9) $M_p = 5\%$
$T_s = 4s$

$N_p = PO\%$

$t_s = \dfrac{4}{\zeta\omega_n} \rightarrow \omega_n = 1{,}54 \text{ rad/s}$

$G = \dfrac{G(s)}{1 + G(s)} = \dfrac{b}{s^2 + cs + d} = \dfrac{K\omega_n^2}{s^2 + 2\zeta\omega_n s + \omega_n^2}$